Date: 14/10/2026

Experiment No. 2

Signal Operations and Transformations

OBJECTIVES

(i) To perform amplitude scaling and time scaling of continuous-time and discrete-time signals and analyze their effects on signal characteristics.

(ii) To perform addition and multiplication of two discrete-time sequences and analyze the resulting signal characteristics.

Software / Laboratory Environment

MATLAB

(i) To perform amplitude scaling and time scaling of continuous-time and discrete-time signals and analyze their effects on signal characteristics.

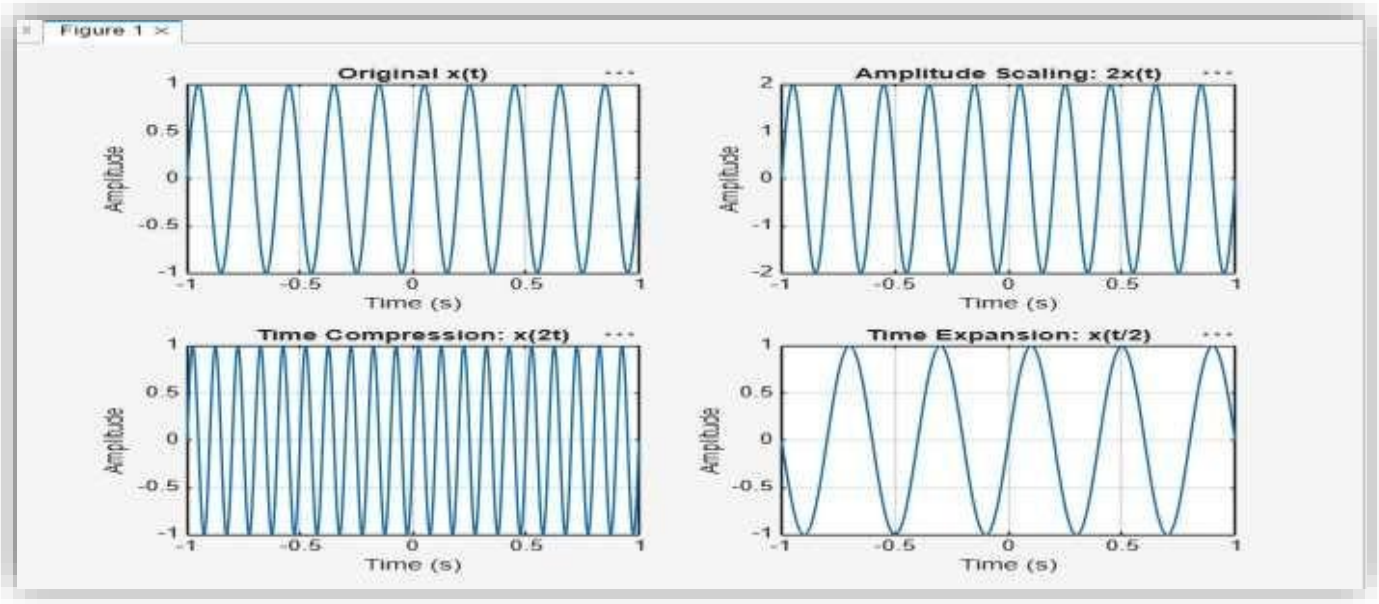
Experimental Procedure

1. Generate the base signal $x(t) = \sin(2\pi 5t)$.
2. Apply amplitude scaling using $A = 2$.
3. Apply time compression using $x(2t)$.
4. Apply time expansion using $x(t/2)$.
5. Plot the original and transformed signals using common axes.
6. Measure the amplitude and period of the transformed signals.
7. Compare the measured results with the theoretical scaling relationships.

/MATLAB Drive/Experiment_2_1.m

```
1   clc; clear; close all;
2   t = -1:0.001:1;
3   x = sin(2*pi*5*t);
4   yA = 2*x;
5   yC = sin(2*pi*5*(2*t));
6   yE = sin(2*pi*5*(t/2));
7   figure;
8   subplot(2,2,1); plot(t,x); grid on;
9   title('Original x(t)'); xlabel('Time (s)'); ylabel('Amplitude');
10  subplot(2,2,2); plot(t,yA); grid on;
11  title('Amplitude Scaling: 2x(t)'); xlabel('Time (s)'); ylabel('Amplitude');
12  subplot(2,2,3); plot(t,yC); grid on;
13  title('Time Compression: x(2t)'); xlabel('Time (s)'); ylabel('Amplitude');
14  subplot(2,2,4); plot(t,yE); grid on;
15  title('Time Expansion: x(t/2)'); xlabel('Time (s)'); ylabel('Amplitude');
```

Code



Output

(i) To perform addition and multiplication of two discrete-time sequences and analyze the resulting signal characteristics

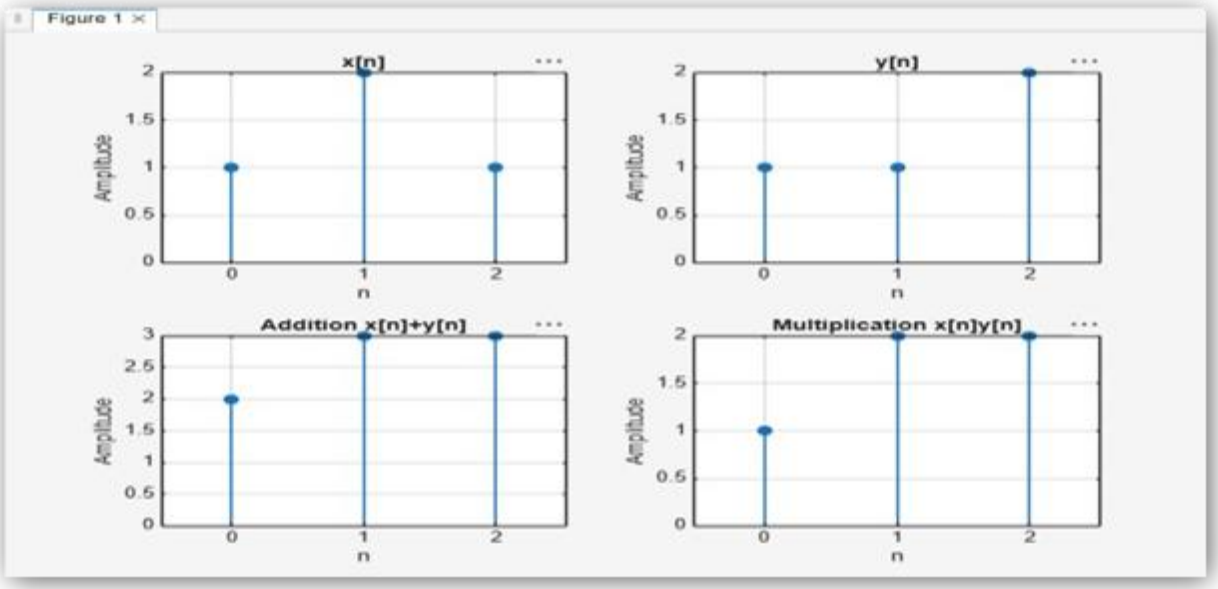
Experimental Procedure

1. Define the sample index n.
2. Enter x[n] and y[n] with aligned indices.
3. Compute the sum sequence.
4. Compute the product sequence using pointwise multiplication.
5. Plot the input and resulting sequences using stem plots.
6. Compare the calculated samples with manual/theoretical results.

/MATLAB Drive/Experiment_2_3.m

```
clc;
clear;
close all;
n = 0:2;
x = [1 2 1];
y = [1 1 2];
zSum = x + y;
zProd = x .* y;
figure;
subplot(2,2,1); stem(n,x,'filled'); grid on;
title('x[n]'); xlabel('n'); ylabel('Amplitude');
subplot(2,2,2); stem(n,y,'filled'); grid on;
title('y[n]'); xlabel('n'); ylabel('Amplitude');
subplot(2,2,3); stem(n,zSum,'filled'); grid on;
title('Addition x[n]+y[n]'); xlabel('n'); ylabel('Amplitude');
subplot(2,2,4); stem(n,zProd,'filled'); grid on;
title('Multiplication x[n]y[n]'); xlabel('n'); ylabel('Amplitude');
```

CODE



OUTPUT

RESULT:

The original and transformed signals were compared, and the observed waveform changes agreed with the corresponding theoretical signal-operation relationships.

RUBRICS:

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage	Marks(10)	Marks obtained
(i) Perform amplitude scaling and time scaling of continuous-time and discrete-time signals and analyze their effects on signal characteristics.	**Amplitude & Time Scaling Implementation (MATLAB)**	Generates the original and scaled versions correctly for CT and DT with correct index handling. Uses plot for CT and stem for DT, with proper time and index vectors. Original and scaled plots are placed in subplots with matching axes, titles, axis labels and grid.	Generates most scaled signals correctly, with minor errors in scaling factors, index handling, axis limits or labelling.	Generates only basic scaled signals, with noticeable errors in code, time or index vectors, or plots, and needs assistance.	Unable to generate the scaled signals or produce meaningful plots.	**25**	2.5	
(i) Perform amplitude scaling and time scaling of continuous-time and discrete-time signals and analyze their effects on signal characteristics.	**Analysis of Scaling Effects**	Correctly explains that amplitude scaling changes amplitude and energy but not duration or period. Explains that time compression shortens	Explains most effects correctly, with minor errors or limited justification.	Gives basic observations on the effects. The explanation is incomplete and needs guidance.	Unable to analyze or explain the effects of scaling.	**25**	2.5	

		duration and period (T/a) and raises frequency, that expansion does the opposite, and that a < 0 reverses time. Explains that DT scaling discards samples and needs interpolation. Supports every point with waveform evidence.						
(iii) Perform addition and multiplication of two discrete-time sequences and analyze the resulting signal characteristics.	**Addition & Multiplication Implementation (MATLAB)**	Adds and multiplies two sequences correctly even when their index ranges differ. Aligns the sequences on a common index vector with zero padding, and uses element-wise operator. Plots both inputs and the results	Performs both operations correctly, with minor errors in index alignment, padding, operators or labelling.	Performs the operations on simple equal-length sequences only, with noticeable errors in code or plots, and needs assistance.	Unable to perform the operations or produce meaningful stem plots.	**25**	2.5	

		with stem, correct n-axis, titles, axis labels and grid.						
(iii) Perform addition and multiplication of two discrete-time sequences and analyze the resulting signal characteristics.	**Analysis of Resulting Signal Characteristics**	Correctly determines the amplitude, the length or range of n, and the energy of the sum and product. Explains the periodicity of the result (for example, the sum is periodic with period equal to the LCM of the two periods). Explains how multiplication acts as gating or modulation. Interprets the results clearly with waveform evidence.	Analyzes most characteristics correctly, with minor errors or limited justification.	Gives basic observations on the resulting signals. The analysis is incomplete and needs guidance.	Unable to analyze or interpret the resulting signals.	**25**	2.5	
					Total	**100**	**10**	